Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Plane Shapes: Their Types, Properties, Similarities and Differences

A plane shape is a flat, closed shape drawn on a flat surface. We can name each plane shape by counting its sides (the straight or curved edges) and its corners (where two sides meet). The corners are also called vertices (one corner is a vertex).

Here are the plane shapes you will meet most often:

• A triangle has 3 sides and 3 vertices.
• A square has 4 equal sides and 4 right angles. All four sides are the same length.
• A rectangle has 4 sides and 4 right angles, but only the two opposite sides are equal.
• A parallelogram has 4 sides, with each pair of opposite sides parallel and equal. Its opposite angles are equal too.
• A trapezium has 4 sides, and only one pair of opposite sides is parallel.
• A circle is the special shape that has no straight sides and no vertices — it is one single curved edge that is always the same distance from its centre.

To compare shapes, look at their similarities and their differences.

Similarity: a square, a rectangle, a parallelogram and a trapezium are all quadrilaterals — each has 4 sides. So they are alike because they all have 4 sides and 4 vertices.

Difference: they differ in how many equal sides they have and how many right angles they have. A square has all 4 sides equal and all 4 angles right; a rectangle has opposite sides equal and all 4 angles right; a parallelogram has opposite sides equal but its angles are not all right angles (unless it is a rectangle); a trapezium has only one pair of parallel sides.

A square is the only shape in this group that is both a rectangle (all right angles) and a shape with all four sides equal.

Let's work through comparing two shapes together.

Suppose someone shows you a square and a parallelogram. Both have 4 sides, so both are quadrilaterals — that is one similarity. But now look closer: the square has all 4 sides equal and all 4 angles are right angles, while the parallelogram has only its opposite sides equal and, usually, no right angles at all. That difference lets you tell them apart.

Now think about a triangle and a trapezium. The triangle has 3 sides and 3 vertices; the trapezium has 4 sides and 4 vertices. Their difference is the number of sides. But they share a feature too: both can be divided into triangles when you draw a diagonal (a straight line joining two opposite vertices).

Practice

1. Name each plane shape described below, then state one property that goes with it.
   a) A shape with 3 sides and 3 vertices.
   b) A shape with 4 sides, all of them equal, and 4 right angles.
   c) A shape with no straight sides and no vertices.
2. A rectangle and a parallelogram both have 4 sides. State one similarity and one difference between them.
3. Which of the following is true of a trapezium?
   A. It has two pairs of parallel sides.
   B. It has exactly one pair of parallel sides.
   C. All four of its sides are equal.
   D. It has no vertices.
4. An octagon is a plane shape with 8 sides. If you draw diagonals from one vertex to all the non-adjacent vertices, how many triangles does the octagon divide into?

Section 2: The Parts of a Circle

A circle is a round plane shape. Every point on its edge is the same distance from a fixed point at its very centre. That fixed point is called the centre of the circle. Around the circle, there are several parts you should be able to name, and each one is a line or region with a special name.

Draw a circle and mark its centre with a dot. From the centre, draw a straight line to any point on the edge. That line is called a radius. A radius is the distance from the centre to the edge. You can draw many radii — every radius of the same circle is equal in length.

Now draw a straight line that passes through the centre and touches the edge at two opposite points. This line is called a diameter. The diameter is twice the radius — it is the longest straight line you can draw inside a circle.

Next, draw a straight line that joins two points on the edge but does not pass through the centre. This line is called a chord. Notice the difference: a diameter is a special chord that passes through the centre. Every diameter is a chord, but not every chord is a diameter.

The remaining parts are made from curves and regions of the circle.

The circumference is the whole distance all the way around the circle — the circle's perimeter. It is the curved boundary of the circle.

A curved part of the circumference between two points is called an arc. A short arc is a minor arc, and a long arc is a major arc.

When a chord cuts across the circle, it splits the circle into two regions. The smaller region between the chord and the arc is called a segment. (The larger region on the other side of the chord is the major segment.)

If you draw two radii to the edge, the slice of the circle between them — shaped like a slice of cake or a piece of pie — is called a sector.

Finally, a straight line that touches the circle at exactly one point and never cuts through it is called a tangent. The tangent meets the circle at just one point, called the point of tangency, and the radius drawn to that point meets the tangent at a right angle.

Example. Draw a circle with its centre marked O. From O, draw a radius to a point A on the edge, and a diameter through O meeting the edge at A and B. Now draw a chord CD that does not pass through O. Shade the slice between the two radii OA and OB — that is a sector. Mark the curved part of the edge between C and D — that is an arc. The smaller region between chord CD and the arc is a segment. Draw a straight line that just touches the circle at point A — that is a tangent. The whole curved boundary of your circle is its circumference.

Check your drawing against these parts: centre (O), radius (OA), diameter (AB), chord (CD), arc, segment, sector, tangent, and circumference. You have now labelled every part of the circle.

Practice

1. Look at a circle you can draw. Name the straight line that joins the centre of the circle to a point on its edge.
2. A straight line passes through the centre of a circle and touches the edge at two opposite points. What is this line called?
3. Which of these two statements is correct: (a) every diameter is a chord, or (b) every chord is a diameter? Explain your answer in one sentence.
4. Draw a circle and mark two points on its edge. Connect them with a chord. Which two regions of the circle does this chord split the circle into, and what is the smaller one called?
5. A straight line touches a circle at exactly one point without cutting through it. What is this line called?

Section 3: Finding the Perimeter of Plane Shapes

The **perimeter** of a plane shape is the total distance all the way around its outside edge. You find it by adding up the lengths of every side. For shapes with equal sides, there is often a quicker way, and for a circle the perimeter has its own name — the **circumference**.

Let's look at each shape you need to be able to measure around.

Square — all four sides are equal, so the perimeter is four times one side.

$$\text{Perimeter of a square} = 4 \times \text{side}$$

Rectangle — opposite sides are equal, so you add the length and breadth, then double.

$$\text{Perimeter of a rectangle} = 2 \times (\text{length} + \text{breadth})$$

Parallelogram — opposite sides are equal, so it works just like the rectangle.

$$\text{Perimeter of a parallelogram} = 2 \times (\text{base} + \text{side})$$

Trapezium — has four sides of (usually) different lengths, so you simply add all four sides.

$$\text{Perimeter of a trapezium} = \text{side}_1 + \text{side}_2 + \text{side}_3 + \text{side}_4$$

Regular polygon — a regular polygon has all its sides equal, so multiply one side by the number of sides.

$$\text{Perimeter} = \text{number of sides} \times \text{length of one side}$$

A regular hexagon has 6 sides, a regular pentagon has 5 sides, and so on.

Circle — the perimeter of a circle is called its **circumference**. It is found using the radius (halfway across) or the diameter (all the way across, so the diameter is twice the radius).

$$\text{Circumference} = 2\pi r \quad \text{or} \quad \pi d$$

where r is the radius, d is the diameter, and $\pi = \frac{22}{7}$ (or 3.14).

Example 1 — Square. Find the perimeter of a square of side 8 cm.

All four sides are equal, so:

$$\text{Perimeter} = 4 \times 8 = 32 \text{ cm}$$

Example 2 — Rectangle. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.

$$\text{Perimeter} = 2 \times (\text{length} + \text{breadth}) = 2 \times (7 + 4) = 2 \times 11 = 22 \text{ cm}$$

Example 3 — Regular pentagon. Find the perimeter of a regular pentagon with sides 5 cm.

A regular pentagon has 5 equal sides, so:

$$\text{Perimeter} = 5 \times 5 = 25 \text{ cm}$$

Example 4 — Parallelogram. Find the perimeter of a parallelogram whose base is 6 cm and side is 5 cm.

$$\text{Perimeter} = 2 \times (\text{base} + \text{side}) = 2 \times (6 + 5) = 2 \times 11 = 22 \text{ cm}$$

Example 5 — Regular hexagon. Find the perimeter of a regular hexagon with sides 5 cm.

A regular hexagon has 6 equal sides, so:

$$\text{Perimeter} = 6 \times 5 = 30 \text{ cm}$$

Example 6 — Circle. Find the circumference of a circle with radius 7 cm. Take $\pi = \frac{22}{7}$.

$$\text{Circumference} = 2\pi r = 2 \times \tfrac{22}{7} \times 7 = 2 \times 22 = 44 \text{ cm}$$

Working backwards — finding the missing side. Sometimes you are given the perimeter and must find a missing side or radius.

Example 7. The circumference of a circle is 22 cm. Find its radius. Take $\pi = \frac{22}{7}$.

$$\text{Circumference} = 2\pi r$$
$$22 = 2 \times \tfrac{22}{7} \times r$$

Divide both sides by 2:

$$11 = \tfrac{22}{7} \times r$$

Multiply both sides by 7:

$$77 = 22r$$

Divide both sides by 22:

$$r = \tfrac{77}{22} = 3.5 \text{ cm}$$

Example 8 — Rectangle with an unknown. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of its sides.

Length $= 2d$, breadth $= d$. The total length of the sides is the perimeter:

$$\text{Perimeter} = 2 \times (\text{length} + \text{breadth}) = 2 \times (2d + d) = 2 \times 3d = 6d$$

Practice

1. Find the perimeter of a square of sides 8 cm.
2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
3. What is the perimeter of a regular pentagon with sides 5 cm?
4. Find the perimeter of a parallelogram whose base is 6 cm and side is 5 cm.
5. Find the perimeter of a regular hexagon with sides 5 cm.
6. The circumference of a circle is 22 cm. Find the radius of the circle. Take $\pi = \frac{22}{7}$.
7. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle.

Check your work

Attempt every question before you read this page.

Plane Shapes: Their Types, Properties, Similarities and Differences

1. a) Triangle — it has 3 sides and 3 vertices. b) Square — it has 4 equal sides and 4 right angles. c) Circle — it has one curved edge and no vertices.
2. Similarity: both are quadrilaterals with 4 sides and 4 vertices. Difference: a rectangle has 4 right angles, while a parallelogram does not (its angles are not all right angles).
3. B. A trapezium has exactly one pair of parallel sides.
4. 6 triangles.

   From any one vertex of a shape, you can draw diagonals to every vertex except itself and its two neighbours. For an 8-sided octagon, that is $8 - 3 = 5$ diagonals from one vertex, which divide the shape into $8 - 2 = 6$ triangles. (A 3-sided triangle gives $3 - 2 = 1$ triangle, a 4-sided square gives $4 - 2 = 2$, and so on.)

The Parts of a Circle

1. radius
2. diameter
3. (a) every diameter is a chord, because a diameter is a straight line joining two points on the edge of the circle — it just happens to pass through the centre. Not every chord passes through the centre, so not every chord is a diameter.
4. The chord splits the circle into two segments — a smaller one and a larger one. The smaller region between the chord and the arc is called a segment (specifically a minor segment).
5. tangent

Finding the Perimeter of Plane Shapes

1. 32 cm

   $4 \times 8 = 32$ cm
2. 22 cm

   $2 \times (7 + 4) = 2 \times 11 = 22$ cm
3. 25 cm

   A regular pentagon has 5 equal sides, so $5 \times 5 = 25$ cm.
4. 22 cm

   $2 \times (6 + 5) = 2 \times 11 = 22$ cm
5. 30 cm

   A regular hexagon has 6 equal sides, so $6 \times 5 = 30$ cm.
6. 3.5 cm

   Circumference $= 2\pi r$, so $22 = 2 \times \frac{22}{7} \times r$. Divide by 2: $11 = \frac{22}{7} r$. Multiply by 7: $77 = 22r$, so $r = \frac{77}{22} = 3.5$ cm.
7. $6d$

   Length $= 2d$, breadth $= d$. Perimeter $= 2 \times (\text{length} + \text{breadth}) = 2 \times (2d + d) = 2 \times 3d = 6d$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-d8e76f1a	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____